

BEYOND THE ALERT

A performance, evidence, and contract framework for acoustic gunshot detection technology.

NCI ANALYSIS / DECISION STANDARD

NCI does not treat gunshot detection as inherently successful or unsuccessful. The relevant question is what a local program adds after the alert enters the system: victim aid, evidence, investigative progress, violence reduction, workload, community effects, and comparative value. Renewal should require local evidence across that full chain.

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Evidence update at a glance

14 studies

The 2026 systematic review identified 14 crime studies; its meta-analysis used 44 effect sizes from 8 studies. ^[1]

1.02

Pooled rate-ratio estimate in the meta-analysis, with a 95% confidence interval of 0.90 to 1.16 - no statistically significant overall relationship with crime. ^[1]

-6.5%

Detroit violent-crime estimate in treated areas; the wild-bootstrap p-value was 0.074, significant at 10% but not the conventional 5% level in that specification. ^[2]

+322%

Detroit increase in gunshot-related calls for service; the study found no significant clearance-rate change and no conclusive travel-time difference. ^[2]

The 2026 evidence update changes the frame

The strongest new evidence is not a single city result. In February 2026, the first systematic review and meta-analysis focused on acoustic gunshot detection and crime identified 14 studies and analyzed 44 effect sizes from 8 studies with sufficient information. The pooled rate-ratio estimate was 1.02, with a 95% confidence interval from 0.90 to 1.16, indicating no statistically significant overall relationship between deployment and crime. The authors also found substantial between-study heterogeneity. ^[1]

Three months later, a peer-reviewed Detroit study reported a 6.5% reduction in violent crime in treated areas. That estimate is important because it differs from much prior literature, but it is also statistically suggestive rather than definitive: the wild-bootstrap p-value was 0.074. The same study found an 8.5% increase in recorded nonviolent crime, a 322% increase in gunshot-related calls, a 10.4% decline in non-gunshot Priority 1 calls, no significant change in clearance rates, and no conclusive travel-time difference. ^[2]

EVIDENCE STANDARD

The meta-analysis and Detroit study should be read together. The pooled evidence does not show a general crime-reduction effect; Detroit suggests that effects can differ by local implementation and context. Neither finding substitutes for a jurisdiction's own outcome, workload, cost, and data-quality evidence.

Do not collapse five different outcomes into one claim

Outcome layer	What the evidence can show	Common analytical error
Detection	Additional probable gunfire events that were not reported through 911.	Treating every alert as a confirmed shooting.
Dispatch and response	Earlier awareness, assignment time, travel time, units sent, and victim-search activity.	Equating a faster alert with a faster arrival or better outcome.
Evidence and investigation	Casings, ballistics, witnesses, firearms, leads, arrests, and case progress attributable to alerts.	Reporting total evidence or arrests without an incremental comparison.
Victim aid	Victims located, EMS activation, first aid, transport, and time to treatment.	Assuming medical benefit without linking the alert to discovery or care.

Outcome layer	What the evidence can show	Common analytical error
Violence and legitimacy	Shootings, victimization, repeat gunfire, enforcement geography, complaints, and community experience.	Treating operational activity as proof of crime reduction or public acceptance.

Across Denver, Milwaukee, and Richmond, an NIJ-supported study found roughly twice as many shooting notifications and two to three times as much staff time after implementation. Response was generally faster for technology alerts, but the time advantage largely occurred between notification and officer assignment rather than during officer travel. ^[3]

Detroit is a useful test, not a national verdict

The Detroit study used a staggered rollout. Areas activated in March 2021 were compared with areas not receiving coverage until October 2022, using police data from January 2017 through October 2022. The design strengthens the analysis, but the period ends before later expansions, operational changes, and current contracts. ^[2]

Detroit finding	Estimate	How to interpret it
Violent crime	6.5% lower	Associated reduction in treated areas; $p=0.074$ in the wild-bootstrap test cited by the authors.
Recorded nonviolent crime	8.5% higher	Authors interpret much of the increase as greater detection of weapons and gunfire rather than more underlying offending.
Gunshot-related calls	322% higher	More awareness and deployments; not equivalent to confirmed shootings.
Other Priority 1 calls	10.4% lower	Operational signal requiring local examination; the study does not prove causal service degradation.
Travel time	No significant difference	No conclusive evidence that officers drove to technology alerts faster than to citizen-reported gunshots.
Clearance rates	No significant change	No detected improvement across crime categories.
Historic contract cost	More than \$70,000 per square mile per year	Cost in contracts analyzed by the study; not a current quote or universal benchmark.

Durham shows why cities need an alert-to-outcome funnel

Durham's 12-month, three-square-mile pilot illustrates how raw alerts narrow as the question becomes more consequential. Researchers counted 1,447 total notifications: 57% technology-only, 15% both technology and 911, and 28% 911-only. Among 282 confirmed gunshot events, the distribution changed to 26% technology-only, 34% both, and 40% 911-only. ^[4]

The pilot recorded 52 serious incidents involving a wound or death. The system alerted for 26, while 911 calls were received for 50. Eight of the missed alerts were attributed to human or system error. Technology-only notifications led to about 2.3 additional deployments per day, seven additional incidents resulting in arrest, and 71 additional evidence-collection incidents. Median response time fell by about 1.2 minutes in the pilot area, but the researchers could not determine that the program reduced gun violence. ^[4]

Stage to publish	Core denominator	Decision question
1. Raw notifications	All vendor and 911 gunfire notifications	How much information entered the system?
2. Corroborated incidents	Events confirmed through audio review, 911, officer observation, evidence, or victim reports	How often did alerts correspond to a real event?
3. Productive responses	Incidents with victim aid, evidence, witness contact, or actionable intelligence	What share produced a documented operational benefit?
4. Case outcomes	Leads, arrests, charges, clearances, or linked investigations attributable to the alert	Did the information advance an investigation?

Stage to publish	Core denominator	Decision question
5. Community outcomes	Shootings, victimization, repeat gunfire, response burden, complaints, and community experience	Did the program improve safety and legitimacy at acceptable cost?

Data quality is a governance requirement

A city cannot evaluate a program if alerts are not consistently coded, corrected, and linked to downstream records. Contract language and agency procedure should establish the record before deployment.

- Use a unique CAD/event code for technology alerts and preserve links to 911 calls, dispatch, incident reports, evidence, EMS, and case records.
- Record the initial alert classification, any human review, later correction, and reason for reclassification without overwriting the original record.
- Define how false alerts, missed incidents, duplicate notifications, indoor gunfire, and events inside vehicles will be handled in performance reporting.
- Preserve the audio, location, analyst notes, version information, and audit logs needed for discovery, evidence review, complaint investigation, and independent evaluation, subject to lawful retention limits.
- Require agency access to event-level data and prohibit contract terms that give a vendor veto over public reporting or unfavorable evaluation findings.
- Document changes in coverage, thresholds, software, review practices, dispatch priority, staffing, and response protocols so apparent outcome changes can be interpreted.

Chicago research published in 2025 did not identify a clear additional racial disparity in arrests or stops beyond patterns associated with standard gunfire calls, but it also showed that both alert pathways were linked to enforcement activity and geography. Community effects therefore need local measurement rather than assumption. ^[7]

Victim discovery and medical response deserve separate measurement

Victim aid can be a legitimate benefit even when crime and clearance effects are uncertain. Camden research examined 267 gunshot-wound cases and reported faster police response and transport than EMS in the observed setting. The study was retrospective and single-center, so it does not establish that the same result will occur elsewhere. ^[6]

A local evaluation should distinguish alerts that only generated a deployment from alerts that materially changed victim discovery, first aid, EMS activation, transport, or time to treatment. These outcomes should not be inferred from alert volume.

A five-question renewal test

1. Incremental information: What events did the system identify that ordinary 911 reporting probably would have missed?
2. Conversion: What share of incremental alerts produced victim aid, evidence, a usable lead, a recovered firearm, an arrest, a clearance, or another documented result?
3. Net system effect: After staffing, unproductive deployments, false or duplicate alerts, and effects on other priority demand, what improved?
4. Legitimacy and distribution: Where were alerts and enforcement concentrated, what complaints or disparities appeared, and what did community feedback show?
5. Comparative value: Would the same money and personnel plausibly produce more safety through another intervention or a different deployment design?

The decision memo should state whether the jurisdiction will continue, modify, narrow, expand, or exit, and tie the decision to metrics defined before the review period. Renewal should be affirmative rather than automatic.

Match the contract to the operating model

Contract area	Minimum requirement
Performance	Definitions for alert, confirmed incident, service level, missed event, correction, and outcome reporting.
Data and audit	Government access to event-level data, audit logs, system versions, analyst changes, and records needed for independent evaluation.
Operational integration	Dispatch priority, units, victim/evidence search, EMS and investigator activation, and alert closure rules.

Contract area	Minimum requirement
Evidence and legal process	Retention, chain of custody, discovery, litigation holds, disclosure, and expert support.
Evaluation	Predeployment baseline, comparison method, reporting schedule, researcher access, and no vendor veto over findings.
Change control	Notice and review before material changes in coverage, thresholds, algorithms, software, analyst practice, or data handling.
Renewal and exit	Performance thresholds, cure or modification process, data export, deletion, hardware removal, and transition assistance.

NIJ's technology-adoption guide begins with problem analysis and separates technical, operational, and governance questions. That structure is useful here: a reliable sensor is not the same thing as a successful program. ^[9]

A practical Michigan reporting standard

- A one-page public program profile: coverage, contract term, full annual cost, retention, sharing, dispatch priority, and evaluation design.
- A quarterly alert funnel: raw notifications, confirmed incidents, productive responses, case outcomes, victim-aid outcomes, and corrections.
- Workload and opportunity cost: units, officer and analyst hours, effects on unrelated high-priority demand, and overtime or staffing needs.
- Community and distribution measures: geography, complaints, enforcement activity, and structured resident feedback.
- A public renewal memorandum using the five-question test and comparing credible alternatives.

Durham's separate sentiment study used nine focus groups and two interviews with 30 residents and found skepticism among many participants. That sample is not representative of every community, but it demonstrates that public experience should be measured rather than treated as an afterthought. ^[10]

Bottom line

The latest research strengthens the case for disciplined local evaluation. The 2026 meta-analysis does not show a general crime-reduction effect, while Detroit reports a context-specific violent-crime estimate alongside a much larger increase in gunshot-related calls and no detected clearance improvement. The program decision should turn on local conversion of alerts into useful outcomes, full operating cost, data quality, community effects, and comparative value.

Selected authorities and current sources

Legislative status and current-source references were checked September 19, 2026. Pending bills, public-comment proceedings, draft standards, and agency programs may change after publication.

- [1] [Huff, Dunlap & Pearson, Does Acoustic Gunshot Detection Technology Reduce Crime? A Systematic Review and Meta-Analysis](#). American Journal of Criminal Justice, published February 6, 2026.
- [2] [Davis & Depew, The Trade-Offs of Gunshot Detection Technology: Evidence from Detroit](#). Journal of Quantitative Criminology, published May 3, 2026.
- [3] [NIJ: Firearm Shootings and the Police Response - three-city GDT study](#). Denver, Milwaukee and Richmond; July 2024.
- [4] [Cook & Soliman, Evaluation of Durham's ShotSpotter Installation](#). Results of a 12-month pilot, released February 2024.
- [5] [NIJ: Learning about ShotSpotter - and Gun Violence - from Chicago](#). Summarizes Chicago and Kansas City evidence.
- [6] [Sandau, Field & Hunter, Detect, Dispatch, Drive](#). Journal of Surgical Research, 2024; Camden gunshot-wound response study.
- [7] [Piza et al., Space-Time Association between GDT Alerts, Calls for Service, and Police Enforcement in Chicago](#). Journal of Quantitative Criminology, 2025.
- [8] [Wilson Center for Science and Justice, ShotSpotter in Durham: A Community Sentiment Evaluation](#). Focus groups and interviews conducted during the Durham pilot.
- [9] [NIJ Criminal Justice Technology Adoption and Implementation Guide](#). April 2025; technical, operational and governance framework.
- [10] [Boehme et al., Gunshot Detection Technology in a Mid-Sized Agency](#). Applied Police Briefings, June 2026; short practice-oriented study.

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